



analysis. Large companies may have access to these tools, but startups often operate with limited budgets.

When simulations are not possible, I stick to standard industry practices. These practices exist for a reason, and following them carefully can still result in a robust system.

Under simulation and analysis, there are a few key things I always focus on:

Perform simulations. If simulation tools are available, I use them to validate critical aspects of the design, such as thermal behaviour, power delivery, and signal integrity.

Perform analysis. Even without simulation tools, I perform detailed analysis on critical components. For example, if a resistor is expected to dissipate one watt of power, I check its derating curve. Every component has a derating characteristic. I look at the temperature at which derating begins and the slope of derating to ensure the component operates safely.

If I am using an LDO, I analyse the expected temperature rise based on power dissipation. For capacitors, I consider DC bias effects. Ceramic capacitors are heavily affected by DC bias. A sixteen-volt-rated capacitor can lose nearly forty per cent of its capacitance even at fifty per cent of the rated voltage. These details matter and directly impact system reliability.

Deep dive into the world of physics. I consciously move away from simplified electrical analogies learned in college, such as water tanks and pipes. Those concepts are useful for basics, but they do not help at higher speeds or in real-world systems.

I need to understand electrical engineering from a wave perspective. Signals on a PCB propagate as electromagnetic waves. Electrons themselves do not move fast enough to explain signal behaviour. It is the wave that propagates through the PCB. Phenomena such as crosstalk, noise, EMI, and EMC issues all exist

because of wave behaviour.

To design better hardware, I focus on physics-based understanding rather than oversimplified models. This shift in thinking significantly improves how I analyse and design systems.

The schematic art: From symbols to footprints

I see schematic and layout design as an art. To create a good schematic and layout, I need to think like an artist. But not a modern abstract artist who randomly places shapes and colours and leaves it to interpretation. My schematic should not look like disconnected components joined with randomly coloured wires. I need to be a traditional artist. That means working with fine strokes, clean structure, and attention to detail. Only then can I create a schematic that is readable, maintainable, and correct.

An artist has a canvas, brushes, and colours. For me, the editor window of my EDA tool is the canvas. The tools provided by the software to place components, connect nets, and add notes are my brushes. The colours of my art are the components themselves.

From a hardware designer's point of view, a component has three elements:

- Symbol
- Footprint
- Parameters

The symbol is used in the schematic. The footprint is used in the layout. The parameters are used in documentation, such as the bill of materials. All three together form a component, and all of them belong to the component library. In large organisations, there may be dedicated librarians who create these components. In smaller organisations, the engineer does everything. Personally, I feel that doing everything is better because it helps me understand the complete design flow rather than just one part of it.

Symbols are the graphical

representation of components. They can represent ICs with multiple pins, MOSFETs, resistors, capacitors, diodes, and many other electrical components. Standard components like resistors, diodes, and MOSFETs usually have well-defined standard symbols. There is no issue there. Real attention is required when creating symbols for ICs.

The tips for creating symbols include:

To pay attention while creating symbols. Every IC is different. I must carefully follow the datasheet and ensure that pin numbers, pin names, and pin functions are exactly as specified. Missing or mislabeling even one pin can spoil the design.

To create functional symbols. Instead of arranging pins numerically, I group pins functionally. Power and ground pins go together. Communication pins go together. Interrupts and outputs are grouped logically. This makes the schematic cleaner and much easier to route and review.

Custom pin numbering for some components. For certain standard components like MOSFETs, pin numbering varies across vendors. One vendor may define pin two as drain, another as source. To avoid confusion, I define my own pin numbering scheme and stick to it consistently.

We must be extremely careful when assigning this numbering to the footprint. The symbol pins and footprint pad numbers must match exactly. Many design failures happen due to incorrect drain and source mapping in packages like SOT 23. These mistakes must be avoided.

While creating footprints, there are several critical points I always follow:

Pay attention to the dimensions. Pad size, pitch, spacing, and overall package dimensions must exactly match the datasheet. These are non-negotiable.

Proper pin sequence. Pin numbering errors between the symbol